

# **From Defect Detection to Flight Clearance: a Bayesian SHM Stack for Reusable Launch Vehicles**

Stage 0–3 implemented and verified on CFRP thermoelastic fields

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**Expendable:** “will it survive *one* flight?” — a single pass/fail margin check.

**Reusable:** “is this vehicle OK to fly *again*, and how many flights remain?” — a recurring, history-dependent decision.

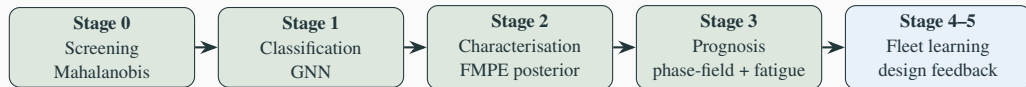
- Same vehicle re-inspected after every flight  $\Rightarrow$  *fixed geometry, fixed individual* is the norm, and its own past-flight scans form a growing healthy reference
- SpaceX (empirically): preventive replacement  $\rightarrow$  *condition-based monitoring*; staged certification 20  $\rightarrow$  40 flights; fleet-leader B1067 at 35 flights (06/2026)

- Cost asymmetry: vehicle + payload loss  $\gg$

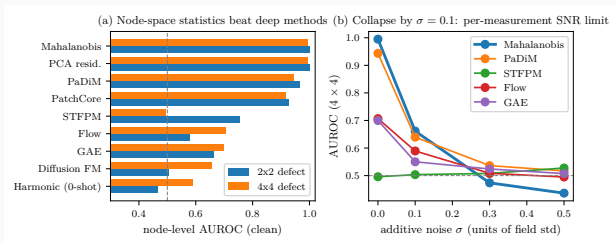
## This work

Formalise that practice as a *measurable, Bayesian, physics-based* pipeline — detect  $\rightarrow$  characterise (with uncertainty)  $\rightarrow$  predict crack growth  $\rightarrow$  clear — applicable from the *first* article, not after a decade of flights.

Data: CFRP interstage thermoelastic full-field (DSPSS, 13 942 nodes) + H3 fairing guided waves.



- **Green = implemented and tested** (2026-06; two git repos, all committed; 19/19 tests green)
- Each stage answers one question: **where?** (Stage 0) → **what?** (Stage 1) → **how bad, with what uncertainty?** (Stage 2) → **fly again?** (Stage 3)
- One data contract end-to-end: posterior  $\theta = (c_x, c_y, \text{layer}, \log_2 \text{size})$  flows from Stage 2 *directly* into Stage 3's defect seeding
- Stages 0–2 are model-light (statistics / amortised inference); Stage 3 carries the mechanics (phase-field fracture)



Score  $s_i = |x_i - \mu_i|/\sigma_i$ , with  $(\mu_i, \sigma_i)$  per node from  $N=2000$  healthy fields. **Node AUROC (clean):**

- Mahalanobis / PCA: **0.999 / 0.995** (2x2 / 4x4)
- PaDiM 0.96, PatchCore 0.93, STTFPM 0.75, GAE 0.66, diffusion 0.50–0.66

## SNR limit (right panel)

All methods collapse by  $\sigma=0.1$ : defect amplitude  $<0.1\sigma$ . Lock-in IR thermography averages  $\sim 10^3$  cycles ( $\sigma \propto 1/\sqrt{K}$ )  $\Rightarrow$  an SNR *requirement*, not a blocker.

## Fixed geometry = template statistics

Re-inspecting the *same* vehicle: healthy manifold is a point cloud around one template.  
Per-node  $(\mu_i, \sigma_i)$  is the correct inductive bias.

## Complementary weaknesses

|                     | Mahalanobis | GNN          |
|---------------------|-------------|--------------|
| unknown defect size | open        | collapses    |
| noise (12%)         | dies        | robust       |
| semantic class      | —           | only option  |
| new airframe        | needs refs  | transferable |

⇒ statistics dominate as a vehicle accumulates flight history; learned models bootstrap *new* airframes and repairs.

Flow-matching posterior estimation, trained on the *existing* FEM set (20k pairs, zero new simulations):

| parameter      | RMSE  | $R^2$            | 90%-coverage |
|----------------|-------|------------------|--------------|
| position $c_x$ | 0.055 | 0.78             | <b>0.944</b> |
| position $c_y$ | 0.021 | 0.83             | <b>0.940</b> |
| size (3-class) | —     | 0.73 (acc. 0.82) | 0.894        |
| layer (depth)  | 2.98  | 0.63             | 0.870        |

- Coverage  $\approx$  target 0.90  $\Rightarrow$  **calibrated** posteriors, safe to propagate downstream
- Depth weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly

Energy  $\Psi = \int \left[ g(d) \psi_e^+ + \frac{G_c}{c_w} \left( \frac{d^2}{\ell} + \ell \nabla d \cdot A \nabla d \right) \right] dV$ , with structural tensor  $A = I + \beta a \otimes a$  (fiber dir.  $a$ ).

**Model** (12/12 tests green):

- multi-ply: per-ply  $A(\theta_{\text{ply}})$ , horizontal bands
- interface bands  $G_c^{\text{int}} / G_c^{\text{ply}} = 0.4$  (DCB/ENF-anchored)  
— pure-PF surrogate of Paggi–Reinoso CZM
- Carrara (2020) fatigue: history  $\bar{\alpha}$  accumulates  $\psi_e^+$  over cycles,  $f(\bar{\alpha})$  degrades  $G_c \Rightarrow$  Wöhler-like growth under *repeated sub-critical* loads

**Verified physics:**

- fiber at  $30^\circ \rightarrow$  crack deflects to  $23.3^\circ$
- low- $G_c$  interface captures crack (73 vs 0 new cells)
- $P(\text{growth})$  monotone in load:  $0 \rightarrow \frac{2}{3} \rightarrow 1$
- FD, plane-strain,  $60 \times 48$  grid,  $\sim 15$  s/run

*Caveat:* load is a normalised Mode-I peel proxy;  $\bar{\alpha}_T$  not yet S–N-anchored.

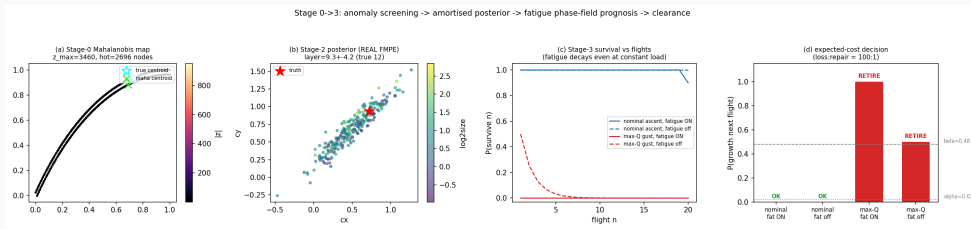
Stage 0→3 run: held-out 4x4 field → Mahalanobis map → *trained* FMPE posterior → fatigue-aware clearance (20-flight horizon).

| load case      | load | fatigue   | $P(\text{grow})$ | $\mathbb{E}[\text{flights}]$ | decision      |
|----------------|------|-----------|------------------|------------------------------|---------------|
| nominal ascent | 0.06 | <b>ON</b> | 0.00             | <b>19.9</b>                  | <b>OK</b>     |
| nominal ascent | 0.06 | off       | 0.00             | 11.0                         | OK            |
| max-Q gust     | 0.10 | <b>ON</b> | 1.00             | 0.0                          | <b>RETIRE</b> |
| max-Q gust     | 0.10 | off       | 0.50             | 1.0                          | <b>RETIRE</b> |

- fatigue *accumulates across flights* —  $P(\text{survive } 20)$  now decays even at constant sub-critical load (no i.i.d. shortcut)
- thresholds from expected cost (vehicle:repair = 100:1  $\Rightarrow \alpha=0.02, \beta=0.48$ )



# End-to-end, one figure: detect $\rightarrow$ characterise $\rightarrow$ clear



Stage 0 anomaly map  $\cdot$  Stage 2 posterior  $\cdot P(\text{survive } n)$  with fatigue  $\cdot$  decision — single held-out CFRP field.

- **Stage 4 — fleet learning:** hierarchical Bayes over tail numbers; formalises the SpaceX fleet-leader practice (lead vehicle = likelihood, fleet = prior update)
- **Stage 5 — design feedback:** fleet damage statistics → layup/reinforcement optimisation (BO)
- **Sim-to-real:** flow-matching bridge ready for first measured full-field data (per-sample normalisation needs no healthy reference)
- **Targets:** JAXA reusable-vehicle programmes; journal paper (anomaly benchmark + calibrated FMPE + fatigue prognosis)

## One sentence

SpaceX learned this loop empirically over a decade — we formalise it so it works from the first inspection of the first article.